

SOP-702: Centrifugal Pump (P-101) Maintenance Standard

Document Ref: SOP-702-REV4 | Target Tag: P-101 | Department: Operations

1. SECTION OVERVIEW & EQUIPMENT SPECIFICATIONS

This manual specifies operational safety and preventive maintenance routines for Centrifugal Water Pump **P-101** installed in primary cooling loop A.

Specification Parameter	Operating Bound	Alert Threshold
Design Flow Rate	450 GPM	Min 380 GPM
Bearing Temperature Limit	75.0 °C	Max Alarm 82.0 °C
Discharge Pressure	120 PSIG	Min 95 PSIG
Recommended Lubricant	Mobil SHC 630	Change Every 2,000 Hrs

2. BEARING LUBRICATION & SEAL INSPECTION PROCEDURE

- Verify oil sump sight glass level is at exactly 50% midpoint.
- Flush contaminated grease via mechanical drain plug until clean lubricant emerges.
- Inspect mechanical face seal for leakage rates exceeding 5 drops per minute.

3. TROUBLESHOOTING & FAILURE DIAGNOSTICS (PAGE 2)

Refer to the decision matrix below when pump P-101 triggers thermal or vibration telemetry alarms.

Observed Fault	Probable Cause	Corrective SOP Action
Bearing Temp > 80°C	Oil Degradation / Contamination	Execute Lubricant Flush SOP-441
Vibration > 3.0 mm/s	Coupling Misalignment / Wear	Re-align Motor Shaft with Laser Tool
Discharge Flow Drop	Impeller Clogging / Cavitation	Inspect Intake Suction Strainer

4. PREVENTIVE MAINTENANCE SIGN-OFF LOG

Authorized Technician: **employee@samiq.ai**
Reviewed by Operations Lead: **admin@samiq.ai**
Date Executed: **2026-07-21**